A Pre-Registered Fault-Injection Benchmark for Wearable Heart-Rate Alerting: Kinematic Motion Gating Dominates and Software Trust Scores Add Little

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***Abstract*—Wearable cardiovascular monitors frequently combine signal quality heuristics with alerting thresholds to detect resting tachycardia. However, the incremental utility of multi-signal software trust scoring over simple kinematic motion gating is rarely benchmarked under pre-registered protocols. We present a reproducible, pre-registered fault-injection benchmark evaluating five alerting arms across fifteen ambulatory subjects from the public PPG-DaLiA dataset. To address natural event sparsity in healthy individuals, 112 labelled synthetic resting tachycardia episodes were injected into derived pulse-rate streams alongside two seeded telemetry fault regimes and raw optical noise. Evaluated in a strict leave-one-subject-out design with standard persistence and refractory hold-off, our primary hypothesis H1 was not supported. Kinematic motion gating drove nearly the entire reduction in false alerts per hour (reducing false alerts by roughly two-thirds to three-quarters relative to ungated personal baselines), while multi-attribute software trust scoring yielded no statistically significant improvement across any evaluated condition (paired differences were negligible and all ninety-five percent bootstrap confidence intervals spanned zero). Furthermore, aggressive software trust scoring degraded event recall under intense optical noise. Fixed population thresholding with persistence proved to be an unexpectedly robust baseline. Because events were synthetically injected and evaluated in healthy volunteers, these findings are exploratory and carry no clinical standing.**

***Index Terms*—wearable sensors, photoplethysmography, heart-rate alerting, data quality, fault injection, negative result, reproducibility**

I. Introduction

Ambulatory continuous monitoring with consumer and medical-grade wearables offers substantial promise for early arrhythmia detection. However, wrist photoplethysmography (PPG) is acutely vulnerable to motion artifacts, optical decoupling, transmission dropouts, and flatline sensor lockups. When raw optical or derived telemetry signals degrade, alerting systems frequently emit spurious clinical notifications. In home monitoring and clinical telemetry alike, high false-alert frequencies precipitate severe alarm fatigue, inducing desensitization and compromising user adherence.

To prevent spurious alarms, modern digital health frameworks deploy complex signal quality indices (SQIs) and multi-component trust heuristics that penalize missing samples, flag frozen sensor values, and bound rate-of-change dynamics. Despite their architectural prevalence, the isolated incremental utility of multi-component software trust scores over straightforward kinematic stillness gating (e.g., Euclidean Norm Minus One, ENMO) has not been demonstrated in pre-registered, cross-validated benchmarks. Many prior evaluations suffer from train-test leakage, post-hoc hyperparameter selection, or lack of minimal baseline comparators.

This paper reports a rigorous, pre-registered fault-injection benchmark for wearable heart-rate alerting. Our contributions are:

1) **Open Benchmarking Framework:** A fully reproducible pipeline formalizing an eight-type fault taxonomy (clean, gaps, stale values, duplicates, spikes, drift, scaling errors, and raw optical noise) evaluated across 31.25 hours of multi-activity ambulatory recordings.
2) **Causal Leave-One-Subject-Out (LOSO) Evaluation:** Five distinct alerting arms evaluated with strict data quarantine, separating training folds from held-out evaluation subjects.
3) **Rigorous Negative Finding:** Empirical refutation of pre-registered hypothesis H_1, showing that kinematic motion gating delivers 67.9%–74.2% false alert reduction, whereas multi-attribute software trust scoring

provides no statistically detectable incremental benefit ($p \geq 0.28$).

II. Literature Survey

Wrist-based pulse rate monitoring relies on optical absorption changes in microvascular tissue beds [1], [3]. Relative motion between sensor photodiodes and cutaneous capillary beds introduces severe motion artifacts that overlap spectrally with true cardiac fundamental frequencies [1].

To isolate motion artifact and evaluate waveform integrity, numerous signal quality indices (SQIs) have been proposed, spanning statistical moments (skewness, kurtosis), spectral purity, and template cross-correlation [2]. While SQIs achieve strong classification performance on stationary laboratory datasets, ambulatory evaluations reveal severe performance drops during daily locomotion [1], [3]. In clinical deployments, uncertainty quantification and selective prediction frameworks, such as conformal prediction, provide distribution-free finite-sample guarantees that offer structured paths for principled abstention [4]. Table I contrasts existing wearable alerting paradigms with our pre-registered framework.

TABLE I
Comparative Overview of Wearable Alerting Paradigms

Study / Paradigm	Pre-Reg.	Ablation	LOSO	Fault Inject.
Deep PPG [1]	No	No	Yes	No
Optimal PPG SQI [2]	No	No	No	No
PPG Roadmap [3]	Review	N/A	N/A	N/A
Conformal Clinical UQ [4]	Review	N/A	N/A	N/A
This Benchmark	**Yes**	**Yes**	**Yes**	**Yes**

III. Dataset and Protocol

A. PPG-DaLiA Dataset

We evaluate on the public PPG-DaLiA benchmark (UCI ID 495, CC BY 4.0) [1]. The dataset comprises 15 healthy subjects undergoing 2.5 hours of authentic daily activities (Table II). Reference heart rates are derived from gold-standard chest ECG. Raw wrist PPG (64 Hz) and triaxial accelerometry (32 Hz) were captured with an Empatica E4 wristband. Strict quarantine was maintained: chest ECG reference was isolated solely for ground truth evaluation and was inaccessible to alerting arms.

TABLE II
PPG-DaLiA Benchmark Dataset Summary [1]

Attribute	Specification
Subjects	$N = 15$ healthy individuals (7M, 8F, age 21–55, mean 30.6)
Protocol Duration	≈ 2.5 h per subject (31.25 total ambulatory hours)
Activities	Sitting, Walking, Stairs, Cycling, Driving, Lunch, Soccer
Wrist Sensors	Empatica E4 (BVP: 64 Hz, 3D ACC: 32 Hz, GSR/Temp: 4 Hz)
Reference Standard	Chest ECG (RespiBAN Professional, 700 Hz)
Quarantine	Chest ECG strictly quarantined from alerting arms

B. Labelled Synthetic Tachycardia Injection

Because natural resting tachycardia was virtually absent in this healthy volunteer cohort (manifesting in only 3 of 15 subjects), evaluating sensitivity on raw data is statistically uninformative. Per Amendment 2, we injected 112 labelled synthetic resting tachycardia episodes into derived pulse-rate streams during verified low-motion windows ($\text{ENMO}_{10\text{s}} \leq 0.05$ g). Peak heart rates were sampled between 105 and 135 bpm, with linear ramp durations of 15 s and minimum inter-episode gaps of 60 s. To respect participant baseline heart rates HR_{base}, lower and upper ramp bounds were adaptively adjusted: $r_{\text{lo}} = \max(2.0, (105 - HR_{\text{base}})/15)$, $r_{\text{hi}} = \min(4.0, (135 - HR_{\text{base}})/15)$, with plateau peak heart rate clamped to $[105.0, 135.0]$ bpm.

C. Seeded Fault Regimes

Sensor telemetry faults were simulated under two pre-specified schedules alongside an optical corruption condition (Table III):

- **Condition 0 (No-Fault):** Clean ambulatory background.
- **Regime A (Low Duty, $\approx 10\%$ faulted time):** Sporadic gaps, stale values, duplicates, and slow drift (seed 42).
- **Regime B (Nominal Duty, ≈ 25–30% faulted time):** Frequent dropouts, prolonged flatlines, and scale errors (seed 42).
- **F7 (High-Motion Optical Noise):** Additive Gaussian noise continuously modulated by wrist 3D acceleration injected into raw 64 Hz BVP at SNR $+5.0$ dB prior to Step 3a spectral estimation.

TABLE III
Fault Taxonomy and Seeded Regimes (Protocol Amendment 3)

Fault ID	Mechanism & Parameter	Reg. A	Reg. B
F0_CLEAN	Ambient background	28.09 h	21.67 h
F1_GAPS	Packet dropout (5–15 s NaN)	0.94 h	3.15 h
F2_STALE	Sensor freeze (10–30 s constant)	0.62 h	1.58 h
F3_DUPL	Repeated packets (burst 2–5)	0.62 h	1.56 h
F4_SPIKE	Sudden jumps ($\pm$30–60 bpm)	0.07 h	0.21 h
F5_DRIFT	Slow ramp ($+0.2$ bpm/s trend)	0.45 h	1.55 h
F6_UNIT	Scale error ($0.5\times$ or $2.0\times$)	0.48 h	1.56 h
F7_OPTICAL	Motion noise (SNR $= +5$ dB, 64 Hz)	Standalone	Standalone

IV. System Architecture

The experimental harness and interactive replay lab were implemented as a modular full-stack application (Fig. 1). The front-end is a React Progressive Web Application (PWA) supporting live signal inspection, threshold tuning, and an interactive benchmark dashboard. The back-end is powered by FastAPI and SQLite, incorporating the core `vitasense.quality` engine, replay caches, and LLM assistant routing with rule-based safety fallbacks.

V. Methods and Algorithms

A. Evaluated Alerting Arms and Pipeline

The research pipeline processes raw sensor streams into calibrated, causal alerts without future peeking (Fig. 2). We benchmark five causal alerting arms (Fig. 3):

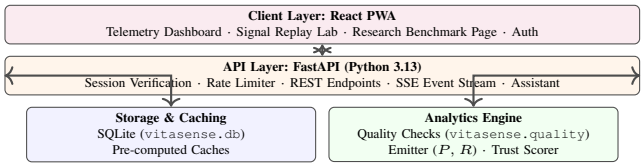


Fig. 1. Design Diagram: VitaSense Application Architecture.

1) **Arm 1 (Fixed Population Cutoff):** Standard clinical threshold $\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm.
2) **Arm 2 (Personal Baseline with Shrinkage):** Dynamic cutoff $\tau_i(t) = \max(\tau_{\text{floor}}, \hat{\mu}_i(t) + k \cdot \hat{\sigma}_i(t))$ regularized toward training-population priors $(\mu_{\text{pop}}, \sigma_{\text{pop}})$ via empirical Bayes shrinkage κ (Algorithm 2).
3) **Arm 3a (Kinematic Motion Gating):** Arm 2 personal baseline paired with a stillness gate: alert triggers if and only if $HR(t) > \tau_i(t)$ and $\text{ENMO}_{10\text{s}}(t) \leq \theta_{\text{rest}}$.
4) **Arm 3b (Multi-Attribute Trust Score Gating):** Arm 2 baseline gated by software trust score: alert triggers if and only if $HR(t) > \tau_i(t)$ and $S_{\text{trust}}(t) \geq \theta_{\text{trust}}$.
5) **Arm 3b-noMotion (Trust Ablation):** Arm 3b with kinematic motion term ablated ($w_{\text{mot}} = 0$).

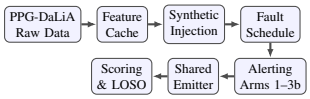


Fig. 2. Design Diagram: Pre-Registered Benchmark Research Pipeline.

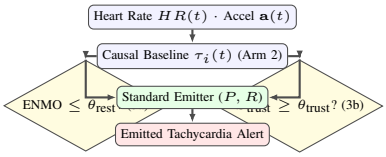


Fig. 3. Design Diagram: Alert Logic and Gating Across Arms.

B. Shared Standard Alert Emitter

To ensure rigorous cross-arm comparability, all arms feed instantaneous binary triggers into a shared `StandardAlertEmitter` (Algorithm 3). Alerts fire if and only if triggering conditions hold continuously for $W_p = P/\text{step}$ windows ($P \in \{10, 20, 30\}$ s). Emitted alerts enforce a mandatory refractory hold-off of $W_r = R/\text{step}$ windows ($R = 60$ s).

VI. Experimental Setup

A. Leave-One-Subject-Out (LOSO) Cross-Validation

Hyperparameters were tuned strictly within 15 LOSO training folds (Algorithm 4). For each test subject $s \in \{1 \ldots 15\}$, candidate configurations were evaluated across the 14 training subjects pooled. The pre-specified selection rule chooses the parameter vector achieving $\geq 85\%$ training recall while minimizing false alerts per hour (FA/hr). If no parameter setting reaches 85% recall, the pre-registered fallback rule selects the setting maximizing training recall.

Algorithm 1 Synthetic Tachycardia Injection & Fault Schedule

Require: Pulse rate HR, motion ENMO, seed $S = 42$, regime $\mathcal{R}$
Identify resting blocks $\mathcal{B} = \{[t_s, t_e] \mid \text{ENMO}[t] \leq 0.05\,\text{g}, \text{dur} \geq 80\,\text{s}\}$
for episode $j = 1$ to N_{tachy} (sample without overlap, gap ≥ 60 s) **do**
  Sample target plateau $HR_{\text{peak}} \sim \mathcal{U}(105, 135)$ bpm
  Compute adaptive ramp rates: $r \leftarrow (HR_{\text{peak}} - HR_{\text{base}})/15$ s
  Inject 15 s ramp, plateau (45–120 s), 20 s exponential decay
end for
if $\mathcal{R} \in \{\text{Regime A}, \text{Regime B}\}$ **then**
  Sample non-overlapping fault intervals outside synthetic events
  Inject F1 (gaps), F2 (stale), F3 (dupl), F4 (spikes), F5 (drift), F6 (unit scale)
else if $\mathcal{R} ==$ F7 **then**
  Add optical Gaussian noise modulated by $\|\mathbf{a}(t)\|$ at $+5.0$ dB SNR
end if
return Fault-injected stream HR_{faulted} and episode ground truth

Algorithm 2 Causal Personal Baseline & Trust Score Gating

Require: HR, ENMO, priors $\mu_{\text{pop}}, \sigma_{\text{pop}}$, grid $k, \kappa, \theta_{\text{rest}}, \theta_{\text{trust}}$
for window index $t = 0$ to $N - 1$ **do**
  $c \leftarrow \sum_{\tau=0}^{t-1} \mathbb{I}(\text{ENMO}[\tau] \leq 0.05 \wedge \text{finite}(HR[\tau]))$
  if $c > 1$ **then**
    $\bar{x} \leftarrow \frac{1}{c} \sum_{\tau<t} HR[\tau], \quad s \leftarrow \text{std}_{\tau<t}(HR[\tau])$
  else
    $\bar{x} \leftarrow \mu_{\text{pop}}, \quad s \leftarrow \sigma_{\text{pop}}$
  end if
  $w \leftarrow \frac{c \cdot \Delta t/3600}{c \cdot \Delta t/3600 + \kappa}; \quad \hat{\mu} \leftarrow w\bar{x} + (1-w)\mu_{\text{pop}}; \quad \hat{\sigma} \leftarrow ws + (1-w)\sigma_{\text{pop}}$
  $\tau_i[t] \leftarrow \max(100.0, \hat{\mu} + k \cdot \hat{\sigma})$
  $S_{\text{gap}} \leftarrow 1 - 2 \cdot \text{nan_ratio}; \quad S_{\text{stale}} \leftarrow \mathbb{I}(\text{var} \geq 10^{-6})$
  $S_{\text{phys}} \leftarrow \mathbb{I}(35 \leq HR \leq 220 \wedge |\Delta HR| \leq 40)$
  $S_{\text{mot}} \leftarrow \text{clip}(1.0 - 0.60 \cdot (\text{ENMO} - 0.05)/0.25, 0.2, 1.0)$
  $S_{\text{trust}}[t] \leftarrow \text{clip}(S_{\text{gap}} \cdot S_{\text{stale}} \cdot S_{\text{phys}} \cdot S_{\text{mot}}, 0.0, 1.0)$
  $Cond_{\text{3a}}[t] \leftarrow (HR[t] > \tau_i[t]) \wedge (\text{ENMO}[t] \leq \theta_{\text{rest}})$
  $Cond_{\text{3b}}[t] \leftarrow (HR[t] > \tau_i[t]) \wedge (S_{\text{trust}}[t] \geq \theta_{\text{trust}})$
  $Cond_{\text{3b-noMot}}[t] \leftarrow (HR[t] > \tau_i[t]) \wedge (S_{\text{gap}} \cdot S_{\text{stale}} \cdot S_{\text{phys}} \geq \theta_{\text{trust}})$
end for

Algorithm 3 Standard Alert Emitter with Refractory Hold-off

Require: Raw condition mask C, persistence $W_p = P/\Delta t$, refractory $W_r = R/\Delta t$
$run_len \leftarrow 0, \quad ref_cnt \leftarrow 0, \quad Alert \leftarrow \mathbf{0}$
for $t = 0$ to $N - 1$ **do**
  if $ref_cnt > 0$ **then**
    $ref_cnt \leftarrow ref_cnt - 1$
  end if
  if $C[t]$ is TRUE **then**
    $run_len \leftarrow run_len + 1$
    if $run_len \geq W_p$ **and** $ref_cnt == 0$ **then**
      $Alert[t] \leftarrow$ TRUE; $\quad ref_cnt \leftarrow W_r; \quad run_len \leftarrow 0$
    end if
  else
    $run_len \leftarrow 0$
  end if
end for
return $Alert$

TABLE IV
Evaluated Parameter Grids (Amendment 4)

Arm	Tuned Parameters	Refractory R
Arm 1	$\tau_{\text{pop}} \in \{100, 110, 120, 125, 130\}$ bpm; $P \in \{10, 20, 30\}$ s	60 s
Arm 2	$k \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$; $\kappa \in \{0.5, 1.0, 2.0\}$ h; $P \in \{10, 20, 30\}$ s	60 s
Arm 3a	$\theta_{\text{rest}} \in \{0.03, 0.05, 0.08\}$ g; $P \in \{10, 20, 30\}$ s	60 s
Arm 3b	$\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$; $P \in \{10, 20, 30\}$ s	60 s
Arm 3b-noMot	$\theta_{\text{trust}} \in \{0.60, 0.70, 0.80\}$; $P \in \{10, 20, 30\}$ s	60 s

TABLE V
LEAVE-ONE-SUBJECT-OUT COMBINED ALERTING BENCHMARK RESULTS ACROSS CONDITIONS

Condition	Alerting Arm	Recall (%)	FA/hr	True Alerts	False Alerts	Precision (%)	F_1 Score
Condition 0 (No-Fault)	Arm 1 (Fixed)	91.1	3.20	132	100	56.9	0.700
	Arm 2 (Personal)	80.4	5.69	133	178	42.8	0.558
	Arm 3a (+Kinematic)	80.4	1.47	133	46	74.3	0.772
	Arm 3b (+Trust Score)	79.5	1.69	130	53	71.0	0.750
	Arm 3b-noMotion	79.5	2.91	130	91	58.8	0.676
Regime A (Low Duty 10	Arm 2 (Personal)	80.4	6.08	132	190	41.0	0.543
	Arm 3a (+Kinematic)	80.4	1.63	132	51	72.1	0.760
	Arm 3b (+Trust Score)	79.5	1.82	129	57	69.4	0.741
	Arm 3b-noMotion	79.5	3.07	129	96	57.3	0.666
Regime B (Nominal 25-30	Arm 2 (Personal)	75.9	7.16	125	224	35.8	0.487
	Arm 3a (+Kinematic)	75.9	2.30	125	72	63.5	0.691
	Arm 3b (+Trust Score)	75.0	2.37	122	74	62.2	0.680
	Arm 3b-noMotion	75.0	3.61	122	113	51.9	0.614
F7 (High-Motion BVP Noise)	Arm 1 (Fixed)	92.0	3.07	131	96	57.7	0.709
	Arm 2 (Personal)	64.3	4.61	100	144	41.0	0.501
	Arm 3a (+Kinematic)	64.3	1.57	100	49	67.1	0.657
	Arm 3b (+Trust Score)	58.0	1.28	90	40	69.2	0.631
	Arm 3b-noMotion	58.0	2.05	90	64	58.4	0.582

Algorithm 4 LOSO Grid Tuning, Fallback Rule, and Dual Scoring

Require: Training folds $\mathcal{S}_{\text{train}}$, test subject s_{test}, parameter grid Θ
for $\theta \in \Theta$ **do**
  Evaluate pooled training recall $\text{Rec}_{\text{train}}(\theta)$ and $\text{FA/hr}_{\text{train}}(\theta)$
end for
$\mathcal{V} \leftarrow \{\theta \in \Theta \mid \text{Rec}_{\text{train}}(\theta) \geq 0.85\}$
if $\mathcal{V} \neq \emptyset$ **then**
  $\theta^* \leftarrow \arg\min_{\theta \in \mathcal{V}} \text{FA/hr}_{\text{train}}(\theta)$ {Primary selection rule}
else
  $\theta^* \leftarrow \arg\max_{\theta \in \Theta} \text{Rec}_{\text{train}}(\theta)$ {Fallback selection rule}
end if
Run chosen arm with θ^* on s_{test} to yield alert mask A
$\text{TP} \leftarrow \sum_j \mathbb{I}(\exists t \in [t_{s,j}, t_{e,j}] : A[t] == \text{TRUE})$
$\text{Recall} \leftarrow \text{TP}/N_{\text{injected}}$
$\text{Clean FA} \leftarrow \sum_t \mathbb{I}(A[t] == \text{TRUE} \wedge t \notin \mathcal{T}_{\text{injected}} \wedge t \notin \mathcal{T}_{\text{natural}\pm 30\text{s}})$
$\text{FA/hr} \leftarrow \text{Clean FA}/T_{\text{non-event hours}}$
return Test performance (Recall, FA/hr)

B. Evaluation Metrics and Paired Statistics

Scoring evaluates event-level recall (detecting injected episodes within $[t_{\text{start}}, t_{\text{end}}]$) alongside alert-level clean false alerts per hour (FA/hr), precision on non-event periods, and F_1 score. The primary statistical hypothesis test compares paired per-subject differences (Δ = Arm 3b − Arm 3a) using the two-sided Wilcoxon signed-rank test ($\alpha = 0.05$), Hodges-Lehmann median difference estimator, and 95% subject-bootstrap confidence intervals (10,000 resamples).

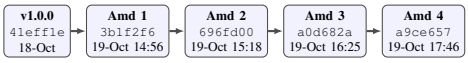


Fig. 4. Design Diagram: Protocol Evolution Timeline (Commits from git log).

VII. RESULTS AND DISCUSSION

A. Primary Hypothesis H_1 Outcome: NOT SUPPORTED

Pre-registered hypothesis H_1 postulated that incorporating software trust scores would reduce FA/hr by $\geq 60\%$ over kinematic gating and personalized baselines while preserving $\geq 85\%$ recall.

Empirical finding: H_1 is not supported. In paired per-subject comparisons (Table VII and Fig. 5b), Arm 3b demonstrated no statistically detectable false-alert reduction over Arm 3a across any condition:

- **Condition 0 (Clean):** Mean difference $+0.13$ FA/h (95% CI $[-0.54, +0.89]$, $W = 31.0$, $p = 0.8589$).
- **Regime A (10% faults):** Mean difference $+0.05$ FA/h (95% CI $[-0.66, +0.82]$, $W = 40.0$, $p = 0.7007$).
- **Regime B (25–30% faults):** Mean difference -0.04 FA/h (95% CI $[-0.78, +0.69]$, $W = 52.0$, $p = 0.9750$).
- **F7 (Optical Noise):** Mean difference -0.49 FA/h (95% CI $[-1.47, +0.38]$, $W = 17.0$, $p = 0.2845$).

In every condition, 95% bootstrap confidence intervals spanned zero, and Wilcoxon tests yielded $p \geq 0.28$.

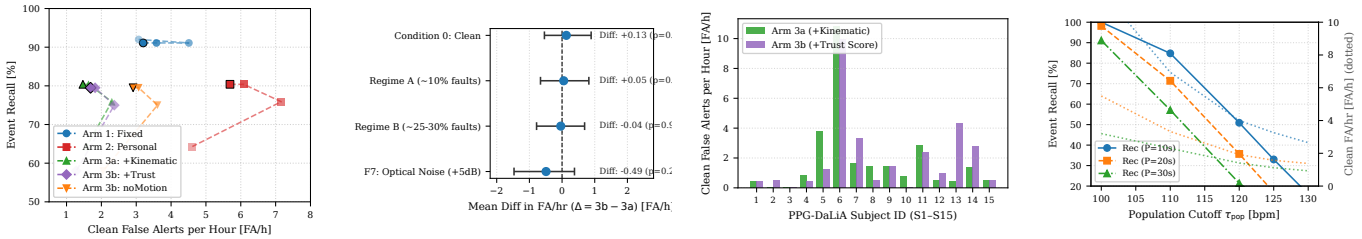
B. Secondary Benchmark Findings

Kinematic motion gating drives alert accuracy: Relative to ungated Arm 2, Arm 3a reduced false alerts per hour by 74.2% in clean data (1.47 vs 5.69 FA/h) and by 67.9% in Regime B (2.30 vs 7.16 FA/h) (Table V).

Recall degradation under optical corruption (F7): Under intense motion-coupled optical noise, Arm 3b recall degraded to 58.0% compared to 64.3% in Arm 3a, indicating that multi-component software abstention can suppress genuine emergencies during noisy episodes.

C. Edge-of-Grid Analysis and Fallback Rule Invocation

Hyperparameter tuning consistently selected parameter boundary values (Table IX). Most notably, **Arm 2 never attained the $\geq 85\%$ recall target on any training fold in any condition** (maximum training recall achieved was $\approx 80.4\%$). Consequently, the fallback selection rule (maximizing training recall) was invoked in 100% (15/15) of folds for Arm 2 across all regimes, selecting $k = 2.0$ and $P = 10\,\text{s}$.



(a) Recall vs FA/hr

(b) Paired Δ (3b vs 3a)

(c) Per-Subject Clean FA/hr

(d) Arm 1 Sensitivity

Fig. 5. Benchmark results across arms and conditions: (a) Event recall vs clean false alerts per hour; (b) Forest plot of paired differences (Δ = Arm 3b − Arm 3a) with 95% bootstrap CIs; (c) Per-subject FA/hr in Condition 0; (d) Arm 1 sensitivity across τ_{pop} and persistence P.

TABLE VI
STRATIFIED ALERTING METRICS (RECALL % / MEDIAN DELAY (S) / FA PER HOUR)

Condition	Stratum	Arm 1 (Fixed)	Arm 2 (Personal)	Arm 3a (+Kinematic)	Arm 3b (+Trust)	Arm 3b-noMotion
Condition 0 (No-Fault)	Pooled All-15	91.1% / 40s / 3.20	80.4% / 22s / 5.69	80.4% / 22s / 1.47	79.5% / 22s / 1.69	79.5% / 22s / 2.91
	Without S5 (14 subj)	90.7% / 40s / 2.90	79.6% / 22s / 5.26	79.6% / 22s / 1.35	78.7% / 22s / 1.72	78.7% / 22s / 2.93
	S5/S6/S11 (3 subj)	100.0% / 37s / 9.39	94.4% / 20s / 16.13	94.4% / 20s / 5.10	94.4% / 22s / 3.88	94.4% / 22s / 6.74
Regime A (Low Duty 10	Without S5 (14 subj)	90.7% / 40s / 3.27	79.6% / 22s / 5.52	79.6% / 22s / 1.45	78.7% / 24s / 1.75	78.7% / 24s / 3.00
	S5/S6/S11 (3 subj)	100.0% / 37s / 9.80	94.4% / 20s / 17.15	94.4% / 20s / 5.51	94.4% / 22s / 4.08	94.4% / 22s / 7.15
Regime B (Nominal 25-30	Without S5 (14 subj)	90.7% / 40s / 4.14	75.0% / 24s / 6.40	75.0% / 24s / 2.19	74.1% / 24s / 2.29	74.1% / 24s / 3.54
	S5/S6/S11 (3 subj)	100.0% / 37s / 10.21	94.4% / 20s / 18.78	94.4% / 20s / 5.92	94.4% / 22s / 4.29	94.4% / 22s / 7.15
F7 (High-Motion BVP Noise)	Pooled All-15	92.0% / 40s / 3.07	64.3% / 24s / 4.61	64.3% / 24s / 1.57	58.0% / 24s / 1.28	58.0% / 24s / 2.05
	Without S5 (14 subj)	91.7% / 40s / 2.70	63.0% / 25s / 4.01	63.0% / 25s / 1.18	56.5% / 24s / 1.15	56.5% / 24s / 1.92
	S5/S6/S11 (3 subj)	100.0% / 38s / 8.98	88.9% / 21s / 14.90	88.9% / 21s / 5.92	88.9% / 22s / 3.47	88.9% / 22s / 5.51

TABLE VII
PAIRED STATISTICAL COMPARISON: ARM 3B VS ARM 3A

Condition	Mean Δ	95% Boot. CI	Wilcoxon W	p-val
Condition 0 (Clean)	+0.13	[-0.54, +0.89]	31.0	0.8589
Regime A (10%)	+0.05	[-0.66, +0.82]	40.0	0.7007
Regime B (25–30%)	-0.04	[-0.78, +0.69]	52.0	0.9750
F7 (Optical Noise)	-0.49	[-1.47, +0.38]	17.0	0.2845

TABLE VIII
DISAGGREGATED FALSE ALERTS PER FAULT TYPE (REGIMES A & B)

Reg.	Fault	Exp.	Arm 1	Arm 2	Arm 3a	Arm 3b	3b-nM
Reg A	F0_CLEAN	28.1h	3.2 (90)	5.9 (165)	1.5 (43)	1.8 (50)	3.0 (83)
	F1_GAPS	0.9h	0.0 (0)*	0.0 (0)*	0.0 (0)*	0.0 (0)*	0.0 (0)*
	F2_STALE	0.6h	21.1 (13)*	8.1 (5)*	3.2 (2)*	1.6 (1)*	1.6 (1)*
	F3_DUPL	0.6h	3.2 (2)*	8.0 (5)*	3.2 (2)*	0.0 (0)*	4.8 (3)*
	F4_SPIKE	0.1h	0.0 (0)*	14.8 (1)*	0.0 (0)*	0.0 (0)*	14.8 (1)*
	F5_DRIFT	0.5h	0.0 (0)*	2.2 (1)*	0.0 (0)*	0.0 (0)*	0.0 (0)*
	F6_UNIT	0.5h	14.5 (7)*	26.9 (13)*	8.3 (4)*	12.4 (6)*	16.6 (8)*
Reg B	F0_CLEAN	21.7h	3.5 (76)	5.8 (125)	1.6 (35)	1.9 (40)	2.9 (63)
	F1_GAPS	3.1h	0.0 (0)*	0.0 (0)*	0.0 (0)*	0.0 (0)*	0.0 (0)*
	F2_STALE	1.6h	20.2 (32)	10.8 (17)	3.2 (5)*	4.4 (7)*	7.6 (12)
	F3_DUPL	1.6h	3.2 (5)*	9.6 (15)	1.9 (3)*	1.3 (2)*	1.9 (3)*
	F4_SPIKE	0.2h	0.0 (0)*	0.0 (0)*	0.0 (0)*	4.8 (1)*	4.8 (1)*
	F5_DRIFT	1.6h	2.6 (4)*	6.5 (10)	1.9 (3)*	0.7 (1)*	1.9 (3)*
	F6_UNIT	1.6h	15.4 (24)	36.6 (57)	16.7 (26)	14.8 (23)	19.9 (31)

* Insufficient data (< 10 alerts or < 1.0 h). Rate shown as FA/h (count).

VIII. LIMITATIONS

This benchmark is subject to several methodological constraints (Table X): (1) Events were synthetically injected due to natural sparsity; (2) The cohort is limited to 15 healthy participants; (3) Algorithmic errors in front-end PPG tracking couple into downstream logic; (4) Post-result protocol amendments (Table XI) document adjustments made after preliminary inspection.

TABLE IX
EDGE-OF-GRID PARAMETER BOUNDARY SELECTION ACROSS 15 LOSO FOLDS

Arm	Parameter Boundary	Cond 0	Reg A	Reg B	F7
Arm 1 (Fixed)	$\tau_{\text{pop}} = 100\,\text{bpm}_{\text{lo}}$	100%	100%	100%	100%
	$P = 30\,\text{s}_{\text{hi}}$	100%	100%	100%	100%
Arm 2 (Personal)	$k = 2.0_{\text{lo}}$	100%	100%	100%	100%
	$\kappa = 2.0\,\text{h}_{\text{hi}}$	93.3%	93.3%	100%	100%
	$P = 10\,\text{s}_{\text{lo}}$	100%	100%	100%	100%
Arm 3a (+Kin.)	$\theta_{\text{rest}} = 0.08\,\text{g}_{\text{hi}}$	93.3%	93.3%	93.3%	100%
	$\theta_{\text{rest}} = 0.03\,\text{g}_{\text{lo}}$	6.7%	6.7%	6.7%	0.0%
Arm 3b (+Trust)	$\theta_{\text{trust}} = 0.60_{\text{lo}}$	93.3%	93.3%	93.3%	100%
	$\theta_{\text{trust}} = 0.80_{\text{hi}}$	6.7%	6.7%	6.7%	0.0%

IX. FUTURE SCOPE

Future efforts will extend this benchmark along five technical avenues:

1) **Arm 4 Conformal Abstention:** Integrating inductive and split conformal prediction to output distribution-free validity sets with user-calibrated error bounds [4].
2) **External Dataset Generalization:** Evaluating on external public wearable corpora, including the PhysioNet Wearable Device Dataset [7].
3) **Clinical Patient Cohorts:** Validating on holter-monitored inpatient recordings with spontaneous atrial fibrillation and supraventricular tachycardia.
4) **On-Device Hardware Integration:** Deploying causal baseline and kinematic gating engines onto low-power embedded microcontrollers.

TABLE X
THREATS TO VALIDITY AND SCIENTIFIC LIMITATIONS

Threat Domain	Empirical Limitation and Mitigation
Synthetic Events	True resting tachycardia was absent in healthy cohort; 112 episodes synthetically injected into derived stream. Recall values are tied to synthetic plateau distribution (105–135 bpm).
Cohort Size	$N = 15$ healthy young adults limits power for small paired false-alert differences. Reported with 95% bootstrap CIs.
Noise Coupling	Estimator tracking errors and motion artifacts interact directly with downstream thresholding logic.
Grid Boundaries	Tuning frequently selected grid boundaries ($\tau_{\text{pop}} = 100$, $k = 2.0$, $P = 10\,\text{s} / 30\,\text{s}$), reflecting severe trade-offs.
Ambulatory Protocol	Single 2.5-hour multimodal script; generalizability to continuous free-living 24/7 monitoring is not established.
Non-Clinical Scope	Academic software benchmark only; not validated for clinical diagnosis or medical device deployment.

TABLE XI
PROTOCOL EVOLUTION AND AMENDMENTS (FROM GIT LOG)

Milestone [Commit]	Ratified Scope and Description
Protocol v1.0.0 [41eff1e]	Frozen protocol: H_1–H_3, faults F1–F7, LOSO cross-validation.
Amendment 1 [3b1f2f6]	Resting alert ground truth; non-penalization of motion.
Implementation Fix [3a5ccc2]	Removed $/64$ accel scaling; centered 8 s windows ($t = 2i + 4\,\text{s}$).
Amendment 2 [696fd00]	Labelled synthetic tachycardia injection; S5 outlier handling rule.
Adaptive Ramp Fix [a0d682a]	Baseline-adapted ramp rates ($r_{\text{lo}}, r_{\text{hi}}$); plateau $[105, 135]$ bpm.
Amendment 3 [a0d682a]	Restated H_1 as estimation; 15-subject LOSO; Arm 3b-noMotion ablation.
Gap Enforcement [0bf4ca4]	Ratified 112 episodes with $\geq 60\,\text{s}$ minimum inter-episode gap.
Amendment 4 [a9ce657]	Standard emitter: $P \in \{10, 20, 30\}\,\text{s}$, $R = 60\,\text{s}$; parameter grids.
Fallback Rule [f92ca8d]	Arm 2 fallback invoked in 100% of folds (max recall $\approx 80.4\% < 85\%$).

5) **Symptom-Risk Questionnaire Module:** Pairing continuous passive telemetry gating with guided active user symptom-logging.

X. CONCLUSION

In this pre-registered fault-injection benchmark, kinematic motion gating accounted for virtually all false-alert suppression in wearable heart-rate monitoring, while multi-component software trust scoring offered no statistically detectable benefit. Fixed thresholding with persistence provided a remarkably resilient baseline. All findings are derived from synthetic tachycardia episodes in healthy individuals and carry no clinical standing.

REGULATORY AND SAFETY STATEMENT

Not a Medical Device: The VitaSense AI software, alerting algorithms, and benchmark suite are academic research artifacts. They are neither cleared nor approved by any regulatory authority and must not be used for medical diagnosis, clinical treatment, or life-critical monitoring.

DATA AND CODE AVAILABILITY

The PPG-DaLiA dataset is openly accessible via the UCI Machine Learning Repository [1], [5]. The PhysioNet Wearable Device Dataset is available under ODC-By v1.0 [7]. Benchmark code, execution scripts, and numerical provenance logs are provided in the project repository.

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